Rayeed Aabir Ahsan

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Personal Information

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Google Scholar : scholar.google.com/citations?user=gg30_ssAAAAJ

Education

North South University, Dhaka, Bangladesh Jan 2022 — Dec 2025
Bachelor of Science in Computer Science and Engineering
CGPA : 3.33 / 4.00; Full Academic Scholarship (100%) as financial aid
Relevant Coursework : Pattern Recognition, Introduction to Robotics, Directed Research, Senior Design Project, Computer Organization and Architecture, Operations Research, Database Management Systems, Probability & Statistics

Standardized Tests

IELTS (Academic) : 7.5 Overall, CEFR C1 (Listening 8.5, Reading 7.5, Writing 6.5, Speaking 7.5); November 2025
GRE (General) : 305 (Verbal 145, Quantitative 160, Analytical Writing 4.5); August 2026

Publications & Submissions

EV Scoping Review : M. S. Sadik, M. Samia, R. Islam, **R. A. Ahsan**, "A scoping review of electric vehicle adoption barriers and policy gaps in South Asia," *Discover Vehicles* (*Springer Nature*), vol. 2, art. 36, 2026. doi:10.1007/s44465-026-00040-1. ***Published.***

AIRC-Fusion : **R. A. Ahsan**, R. Islam, F. R. Hridi, M. C. Prianka, and R. M. Rahman, "Real-Time Infrared and Visible Video Fusion on Processor-Only Edge Hardware Using Vectorized Direct Mapping," Cureus Journal of Computer Science, vol. 3, no. 1, 2026, doi: 10.7759/s44389-026-00341-2. ***Published.***

Decolorization : **R. A. Ahsan**, R. M. Rahman, "Per-Image Cue Weighting for Real-Time Decolorization on Edge Devices." ***Manuscript under review at IEEE Access.***

STARL : **R. A. Ahsan**, R. Islam, J. Mohsin, S. Ahmed, "A Leakage-free Benchmark for Continual Learning in Medical Imaging: What is Forgotten, Where, and at What Clinical Cost." ***Manuscript prepared for Medical Image Analysis (Elsevier).***

Research Experience

Scoping Review: EV Adoption Barriers and Policy Gaps in South Asia Apr 2026 — Aug 2026
Co-Author (4th of 4)
Lead Author : MD Shiyan Sadik, Department of Environmental Science & Management, NSU
Responsibilities : Literature collection and source verification; wrote the Conclusion; managed referencing and citations; proofreading
Achievements : PRISMA-ScR review published in Discover Vehicles (Springer Nature), Sept 2026

AIRC-Fusion: Real-Time IR–Visible Video Fusion on CPU-Only Edge Hardware Jan 2025 — Aug 2026
First Author & Corresponding Author
Supervisor : Dr. Rashedur M. Rahman, Professor, Department of Electrical and Computer Engineering, NSU
Responsibilities : Concept and vectorized reformulation; all code and experiments (9 methods, 5 video sequences); Raspberry Pi 5 live system; manuscript writing, revisions and journal correspondence; presenter
Achievements : 46–106× faster than the original implementation with identical output; 51.2 FPS live on Raspberry Pi 5 while all 8 baselines stayed under 10 FPS; new optical-flow-free temporal instability score; open code and Zenodo archive, published in Cureus Journal of Computer Science, Oct 2026

Per-Image Cue Weighting for Real-Time Decolorization on Edge Devices July 2026 — Present
First Author (Research Assistant project)
Supervisor : Dr. Rashedur M. Rahman, Professor, Department of Electrical and Computer Engineering, NSU
Responsibilities : Led every stage independently: method design and 155k-parameter ViT selector, experiments on five datasets against nine baselines, Raspberry Pi 5 deployment with power and thermal logging, IEEE-format manuscript
Achievements : Highest near-iso-luminant contrast score of 10 methods on Color250 (E = 0.8398); reaching 22.9% over BT.709 across four datasets; 12.72ms/frame on one CPU thread (~7× faster than OpenCV's *cv2decolor*); 12–26 FPS live on Raspberry Pi 5

STARL: Leakage-Free Continual-Learning Benchmark for Medical Imaging Jan 2025 — Present

First Author

Supervisor : Ms. Silvia Ahmed, Senior Lecturer, Department of Electrical and Computer Engineering, NSU

Responsibilities : Study design; complete pipeline (data integrity, training, evaluation); all experiments (12 architectures, 3 seeds, 2 external test sets); analysis; manuscript writing

Achievements : Perceptual-hash de-duplication removed 2,099 near-duplicates (9.2%) before splitting; replay beat naive fine-tuning in 14/14 paired runs (89% of the joint upper bound); showed forgetting tracks task similarity, not class count; 5–11× compression by distillation with no accuracy loss

Professional Experience

Research Assistant July 2026 — Present

Department of Electrical and Computer Engineering, North South University

Supervisor : Dr. Rashedur M. Rahman, Professor

Responsibilities : Lead researcher on a project supported by NSU Faculty Research Grant CTRG-25-SEPS-06 (method design, experiments, *Raspberry Pi 5* deployment); first-author manuscript under review at *IEEE Access*

Teacher (Part-Time) Jan 2023 — Apr 2024

Summerfield International School, Dhaka

Responsibilities : Taught ESL, Mathematics, Physics, Chemistry, Economics and Accounting; Editorial Club supervisor; event organizing and coordination; student support

Achievements : Project lead for the school's *50th Anniversary Magazine*; compiled alumni database; created quarterly newsletters; hosted quiz competitions

Freelance Data Analyst Jan 2022 — Dec 2022

Connectify Inc. (New York, USA; remote)

Responsibilities : Processed, cleaned and verified patient medical and insurance records in Excel databases and the client's billing platform, maintaining accuracy and confidentiality

Technical Skills

Languages : Python, C++, HTML, CSS, JavaScript, SQL, R

Frameworks : PyTorch, Django

ML/DL Libraries : NumPy, OpenCV, PyWavelets, scikit-learn, Pandas, TensorFlow

Tools and Platforms : VSCode, Git & GitHub, Zenodo (DOI archiving), Kaggle (T4/P100 GPUs), Google Colab

Hardware : Raspberry Pi 5 (live camera pipelines, CPU-only deployment, power/thermal logging)

Research Methods : Benchmark design, ablation studies, statistical testing (Wilcoxon, paired t-test, bootstrap CIs), representation analysis (CKA), knowledge distillation, data-leakage control (perceptual hashing)

Projects

University Website Navigator : Scraped 100k+ URLs of utoronto.ca domain for LLM-powered navigation (ongoing)

Personal AI Assistant : Local multimodal LLM system for automation (ongoing)

Personal Website : Research website hosted on GitHub Pages (hosted at rayeedaabir.github.io)

Awards & Certifications

ALOHA International Competition : Bronze (Malaysia, 2011)

Daily Star Awards : 2019, 2021

NSU Academic Scholarship : Full (100%)

National Mathematics Olympiad : Bronze (2014)

Edexcel High Achievers Award : 2019

IAAC Finalist : 2025

Extracurriculars & Leadership Roles

Editorial Club Supervisor; 50th Anniversary Organizing Committee; mentoring and tutoring

Research Interests

- Algorithmic Optimization
- Color Imaging & Decolorization
- Edge AI & Resource-Constrained Systems
- Image Processing & Image Fusion
- Lightweight Vision Transformers
- Benchmarking & Evaluation Rigor
- Continual Learning
- Efficient & Real-Time Computer Vision
- Knowledge Distillation & Model Compression
- Medical Image Analysis